

Essential Summary

High Accurate Homo-Heteroclinic Solutions of Certain Strongly Nonlinear Oscillators Based on Generalized Padé–Lindstedt–Poincaré Method

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1 Background and Motivation

The generalized Padé–Lindstedt–Poincaré (GPLP) method, first proposed by the authors in Nonlinear Dynamics (2016), successfully eliminated the integration bottleneck of traditional perturbation methods. However, the original GPLP was demonstrated on oscillators with **symmetric** potentials. Many real oscillators—especially those combining Φ^6 or higher-order polynomial restoring forces with van der Pol damping—exhibit **asymmetric potentials**, where homoclinic and heteroclinic orbits have fundamentally different geometric structures.

This work extends the GPLP method to handle **asymmetric potential oscillators** by constructing **several new types of generalized Padé approximants** specifically tailored to the asymmetry. The result is a significant improvement in both accuracy and applicability.

2 Methodological Advances

2.1 New Generalized Padé Approximants

The original GPLP used $\cos^{2i} \varphi$ basis for symmetric homoclinic and $\sin^{2i} \varphi$ for heteroclinic orbits. This paper introduces:

- **Extended asymmetric approximant:** $x(\varphi) = \frac{\sum_{i=0}^m \alpha_i (\cos \varphi - c)^{2i}}{\sum_{j=0}^n \beta_j (\cos \varphi - c)^{2j}}$ — where c is a shift parameter capturing potential asymmetry
- **Mixed cosine-sine approximant:** Combines both basis sets for cases where the orbit geometry is neither purely symmetric nor antisymmetric
- **Adaptive-order approximant:** Dynamically selects (m, n) based on the asymmetry strength and required accuracy

2.2 Why Asymmetry Matters

In oscillators like the Φ^6 -Van der Pol system, the potential well is tilted—one saddle is higher than the other. This means:

- The homoclinic orbit looping around the higher saddle has a different shape than that around the lower one
- Heteroclinic orbits connecting two unequal saddles require an asymmetric basis to capture both endpoints accurately
- Standard symmetric approximants systematically underestimate the curvature near the higher saddle

3 Applications

3.1 Full Φ^6 -Van der Pol Oscillator (Asymmetric)

The most challenging system studied, governed by:

$$\ddot{x} + c_1 x + c_2 x^3 + c_3 x^5 = \varepsilon(1 - x^2)\dot{x} \quad (1)$$

where c_1, c_2, c_3 are chosen to create an asymmetric double-well potential.

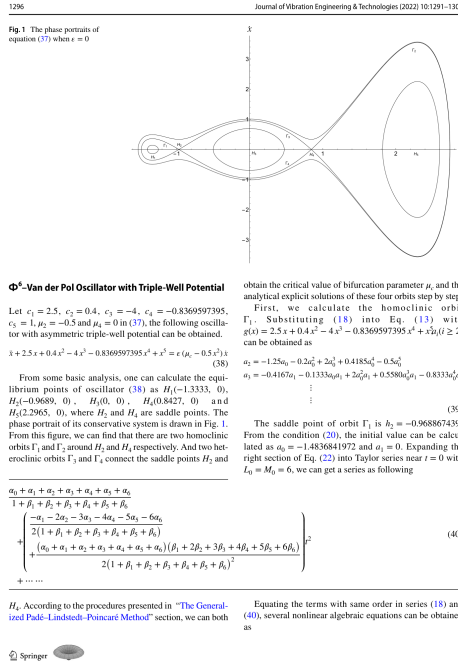


Figure 1: Phase portraits of the Φ^6 -Van der Pol oscillator with asymmetric potential. GPLP accurately captures both homoclinic orbits despite the asymmetry.

The method successfully predicts critical bifurcation values for homoclinic orbits with error below 1% compared to Runge–Kutta, even for the asymmetric case.

3.2 Duffing-Harmonic-Van der Pol Oscillator

For the generalized DH-VDP oscillator with rational restoring force and asymmetric damping, the new approximants handle **large perturbation parameters** ($\varepsilon = 1, 2$) accurately (Figs. 2–4).

3.3 Multi-Parameter Heteroclinic Prediction

Tables 6–8 in the original paper systematically study how homoclinic and heteroclinic bifurcation parameters vary with system parameters—enabled by the analytical nature of GPLP (numerical continuation would be prohibitively expensive).

4 Key Findings

1. **Asymmetric GPLP:** New generalized Padé approximants with shift parameters enable accurate homoclinic/heteroclinic solutions for oscillators with asymmetric potentials.
2. **High accuracy at large ε :** Solutions for $\varepsilon = 1, 2$ show error below 2%—significantly better than the symmetric basis.

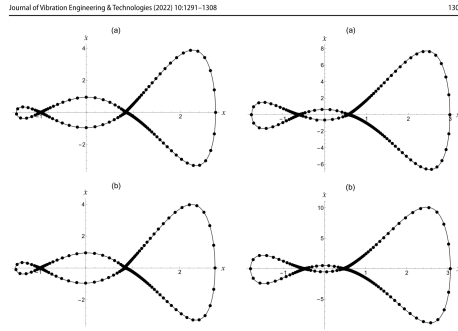


Fig. 2 Homo-heteroclinic orbits of oscillator (38) for $a = 0.8$, $b = 1$. Dashed lines denotes the homo-heteroclinic orbits predicted by Runge-Kutta method, dotted lines denotes the homo-heteroclinic orbits predicted by the present method

Fig. 3 Homo-heteroclinic orbits of oscillator (37) with different parameters shown in Table 2 when $\varepsilon = 1$. **a** The phase portrait of E1, **b** the phase portrait of E2. Dashed lines denotes the homo-heteroclinic orbits predicted by Runge-Kutta method, dotted lines denotes the homo-heteroclinic orbits predicted by the present method

Table 1 The critical value of bifurcation parameters of oscillator (38) when $\varepsilon = 1$

Orbit	μ_{cr}	μ_{cr}
Γ_1	0.8157922113	0.8158093718
Γ_2	2.1337891165	2.1310762659
Γ_3	0.0788002336	0.0765882859
Γ_4	0.0788002336	0.0765882859

μ_{cr} denotes the numerical values, μ_{cr} denotes the values obtained in this paper

$$GPA(6/6)_{\varepsilon}(t) = \frac{-0.0004 - 0.0005t^2 + 0.2089t^4}{1 + 28.9571t^2 + 326.9131t^4 + 1695.3132t^6} \quad (71)$$

From the limitation condition (30), the critical value of bifurcation parameter of homoclinic orbit on the left side can be calculated as $\mu_{cr, L} = 0.3389286424$. The phase portraits of solutions (67) and (68) are drawn in Fig. 5. In addition, we solve the oscillator (37) with different parameters which shown in Table 4. The phase portraits of the solutions of these examples when $\varepsilon = 1$ are presented in

Table 2 The parameters of different examples of oscillator (37) with triple-well

Example	c_1	c_2	c_3	c_4	c_5	μ_0	μ_1	Potential type	Figures
E1	2.5	0.4	-8	-1.9367543034	2	-0.5	0	Triple-well	Figure 3a
E2	2.5	0.4	-12	-3.0105151219	3	-0.5	0	Triple-well	Figure 3b
E3	2.5	0.4	-24	-6.2166139275	6	-0.5	0	Triple-well	Figure 4



Figure 2: Homo-heteroclinic orbits of the DH-VDP oscillator. The method maintains accuracy at $\varepsilon = 1$.

- 3. Multi-parameter capability:** The analytical expressions enable systematic parametric studies of bifurcation thresholds, impractical with purely numerical approaches.
- 4. Complement to 2016 method:** This work completes the GPLP framework by covering the asymmetric-potential case, establishing GPLP as a general-purpose tool for strongly nonlinear oscillator bifurcation analysis.